

Practice Problem Sheet-2

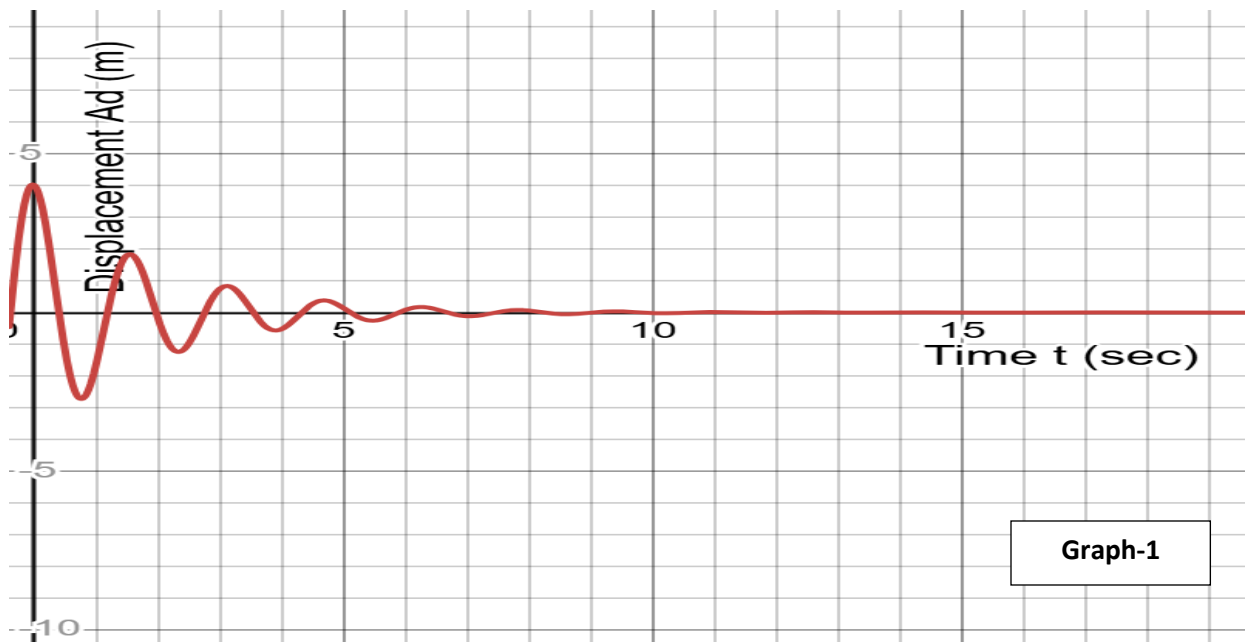
Course Code: PHY 2105/PHY 105 Summer: 2026 [262]

Course Title: Physics Content: DHM+Progressive Wave+Wave Equation

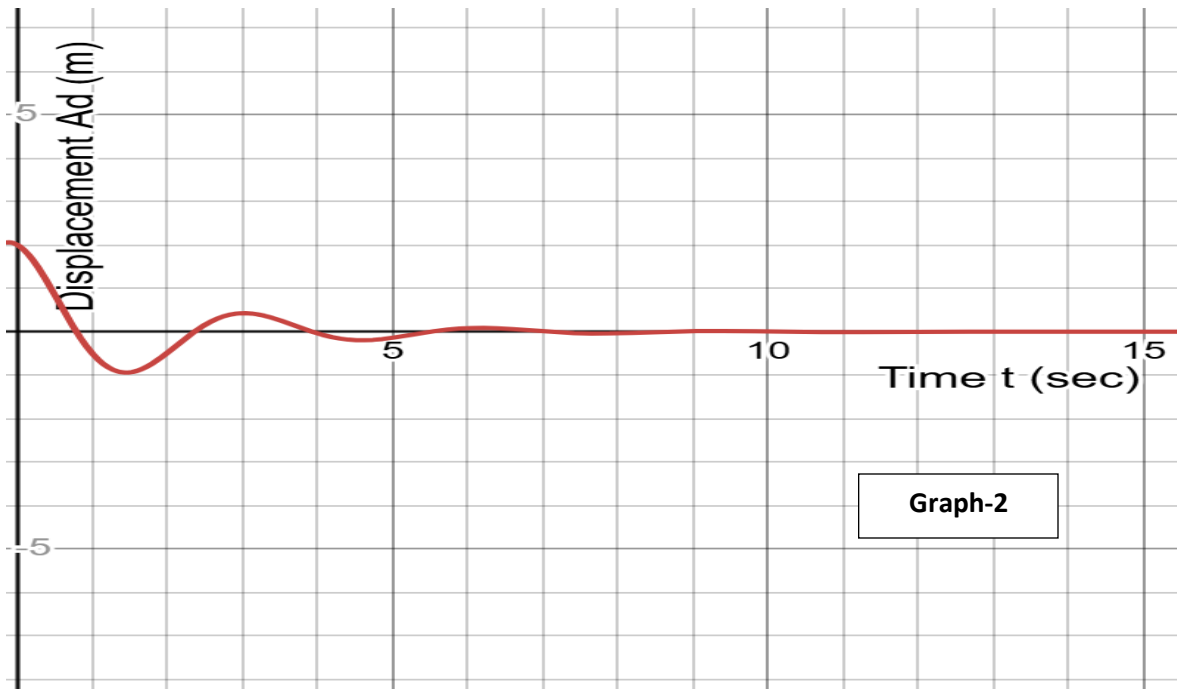
1. The equation of a travelling wave is $y = 4.0\sin(0.10x - 2t)$. Find (i) amplitude, (ii) wavelength, (iii) speed, and (iv) frequency of the wave.
2. When a simple harmonic motion (SHM) is propagated through a medium, the displacement of the particle at any instant of time is given by $y = 5.0\sin \pi(360t - 0.15x)$. Calculate (i) the amplitude of the vibrating particle, (ii) wave velocity, (iii) wave length, (iv) frequency (v) maximum velocity, and (vi) time period.
3. The equation of a progressive wave is given by, $y = 5\sin(100\pi t - 0.4\pi x)$. Calculate the (i) amplitude, (ii) wave length, (iii) frequency, (iv) time period, (v) wave velocity, (vi) angular frequency, (vii) maximum velocity, (viii) maximum acceleration, (viii) phase velocity, and (viii) instantaneous velocity, at $t=3$ s and $x=2$ m.
4. A mobile phone tower transmits a wave signal of frequency 900 MHz. Calculate the length of the waves transmitted from the mobile phone tower.
5. For a travelling wave the displacement is $y = 5\sin 30\pi [t - (x/240)]$. Find the frequency of the wave.
6. A wave from SHM of amplitude +8units travels a line of particles in the direction of positive X axis. At any instant for a particle at a distance of 10 cm from the origin, the displacement is +6 units and at a distance a particle from the origin is 25 cm, the displacement is +4units. Calculate the wavelength.
7. Draw displacement vs. time graph of a DHM for $\omega/\gamma = 10$, $\omega/\gamma = 0.5$, and $\omega/\gamma = 0.03$.
8. A block of mass 1 kg attached to a spring is made to oscillate with an initial amplitude of 12 cm. After 2 minutes the amplitude decreases to 6 cm. Determine the value of the damping constant for this motion.
9. For a damped oscillator $m = 200$ gm, $k = 80$ N/m and $b = 65$ gm/s. (i) What is the period of the motion? (ii) Which type of oscillation it is? (iii) How long does it take for the amplitude of the damped oscillations to drop to half its initial value? (iv) How many oscillations does it complete in life time? (v) How long does it take for the mechanical energy to drop to one-half its initial value? (vi) What is its life time or relaxation time? (vii) The maximum displacement of an undamped oscillator is 35 cm. If the damping is stopped after 20 cycles, What is the damping energy? What is the (viii) ratio of the oscillation amplitude to the initial oscillation amplitude at this cycle? (ix) damping angular frequency? (x) damping frequency? (xi) damping amplitude? (xii) damping energy at $t=3.5$ s? (xiii) damping amplitude at $t=3$ s? (xiv) displacement equation? (xv) damping displacement at $t=4$ sec? (xvi) damping period? (xvii) natural frequency? (xviii) damping factor? (xix) displacement of the particle after 2 cycle? (xx) damping force if $v=6$ m/s? and (xxi) what is the critical damping co-efficient?
10. The displacement of a particle with mass 500 gm following DHM is $y = 6e^{-0.07t} \cos(4\pi t + \delta)$. What is the (i) damping constant? (ii) damping frequency? (iii) natural frequency? (iv) damping displacement at $t=2$ sec? (v) damping energy at $t=5$ sec? and (vi) resonance frequency?
11. Draw the DHM for displacement vs time graph : $y(t) = 5e^{-\alpha t} \cos(6t + \phi)$ up to 2 cycles.

12. If the amplitude of a damped oscillator becomes $1/27$ th of its initial value after 6 minutes, what was the amplitude after 2 minutes?
13. In oscillatory circuit $L = 0.4 \text{ h}$, $C = 0.0020 \mu\text{F}$. (i) What is the maximum value of resistance (R) for the circuit to be oscillatory? and (ii) What is its resonant frequency?
14. Find whether the discharge of capacitor through the following inductive series circuit is oscillatory or not. Given, $C = 0.1 \mu\text{F}$, $L = 10 \text{ mh}$, and $R = 200 \Omega$. (i) If oscillatory, find the frequency of oscillation and resonant frequency. (ii) If it is parallel circuit, then find out the similar characteristics of that circuit, and (iii) If 200Ω is removed from the circuit, then what will be the resistance required for getting a critical damping C_d ?
15. Find whether the discharge of capacitor through the following inductive series circuit is oscillatory or not? Given, $C = 0.2 \text{ nF}$, $L = 10 \text{ mH}$, and $R = 300 \Omega$. If oscillatory, find the frequency of oscillation and resonant frequency. If it is parallel circuit, then find out the similar characteristics of that circuit.
16. Find whether the discharge of capacitor through the following inductive series circuit is oscillatory or not. Given, $C = 1 \mu\text{F}$, $L = 10 \text{ mh}$, and $R = 400 \Omega$. (i) If oscillatory, find the frequency of oscillation and resonant frequency, (iii) If 400Ω is removed from the circuit, then what will be the resistance required for getting a critical damping C_d ? (iv) What will be the critical resistance for parallel circuit?
17. Find whether the discharge of capacitor through the following inductive circuit is oscillatory or not. Given, $C = 0.1 \mu\text{F}$, $L = 10 \text{ mh}$, and $R = 200 \Omega$. (i) If Oscillatory, find the frequency of oscillation. (ii) Also find out the resonance frequency. (iii) What is the displacement equation for this parameters? (iv) What is its displacement at $t = 2 \text{ ms}$?
18. Given $L = 10 \text{ mh}$, $C = 1.0 \mu\text{F}$ and $R = 0.1 \Omega$. How long a time will the charge oscillations decay to half amplitude? What is the damped angular frequency of damped harmonic oscillation?
19. In a spring mass system, the block has a mass of 1.50 kg and the spring constant is 8.00 N/m and $b = 230 \text{ g/s}$. The block is pulled down 12.0 cm and released. (i) What is the amplitude of the damped oscillations at the end of 10 oscillations? and (ii) Find the total energy at that time.
20. A wave has an angular frequency of 110 rad/s and a wavelength of 1.80 m . Calculate (i) the wave number and (ii) the speed of the wave.
21. A transverse wave travels along a medium. The time for a particular point to move from a maximum placement to zero is 0.170 s . The wavelength is 1.40 m and amplitude is 0.25 m . What are the (i) period? (ii) frequency? (iii) wave speed? and (iv) maximum speed of a particle of the medium.
22. Equation of a travelling wave is given by $y = 0.3 \sin(27.1x - 2.72t) \text{ m}$. (i) What is the particle velocity of the medium at $x = 22.5 \text{ cm}$ and at the time $t = 18.9 \text{ s}$? and (ii) What is the maximum speed of the particle at origin?
23. A wave traveling in the positive x direction has a frequency of $f = 24 \text{ Hz}$ and wave length 20 cm . If the amplitude of the wave is 18 cm , (i) draw the wave and (ii) find out the speed of the wave.
24. Find the displacement of an air particle from origin of disturbance at $t = 0.55 \text{ s}$ when a wave of amplitude 0.2 mm and frequency 500 Hz travels along it with a velocity of 350 m/s .
25. A sinusoidal wave of frequency 500 Hz has a speed of 350 m/s . (i) How far apart are two points that differ in phase by $\pi/3 \text{ rad}$? (ii) What is the phase difference between two displacements at a certain point at times 1.00 m apart?
26. A transverse wave of frequency 50 Hz has a speed of 250 m/s is propagating $+x$ axis. If amplitude of wave is 25 cm , find out the maximum speed of the particle at origin.

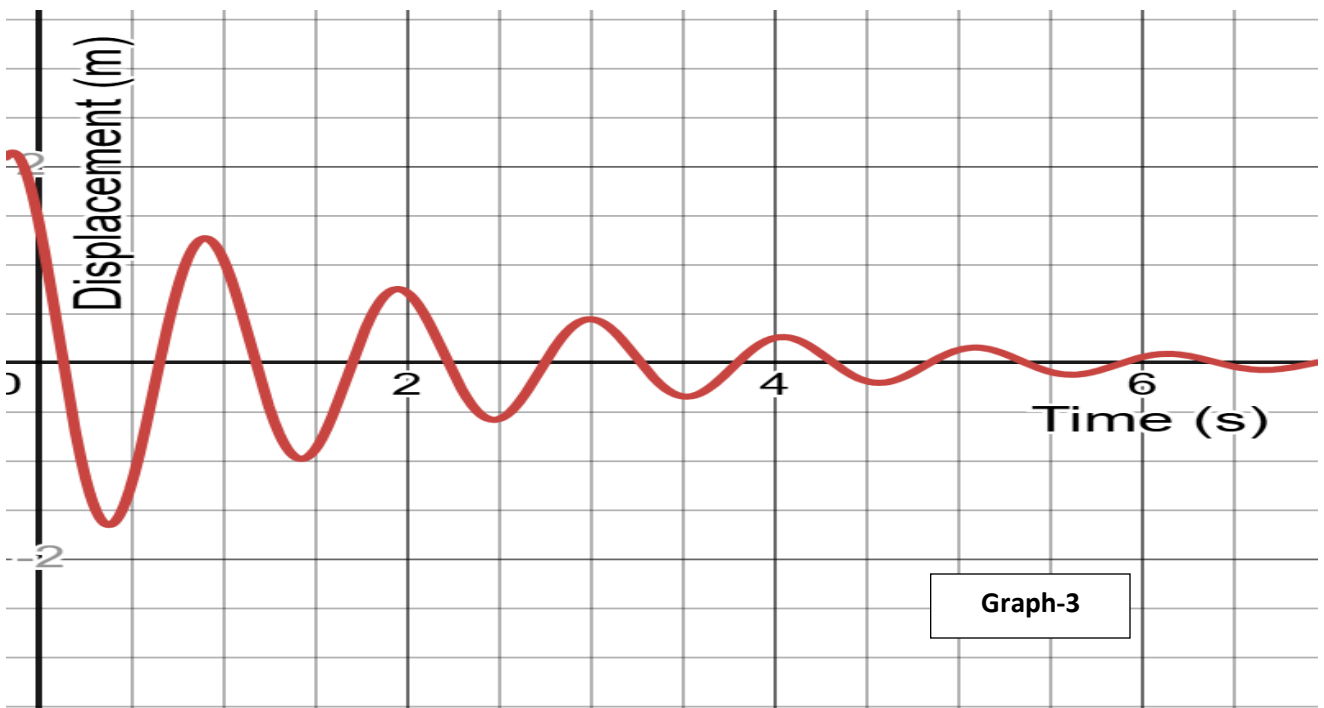
27. A damped harmonic oscillation of a body of mass 0.3 kg is represented by the following equation:
 $x(t) = Ae^{0.5t} \cos(3\pi t - \frac{\pi}{6})$. Find (i) damping coefficient, (ii) natural frequency, (iii) force constant, and (iv) damping factor (damping term) at $t = 2$ ms.
28. The maximum displacement of an undamped oscillator is 20 cm. The damped oscillator has mass $m = 680$ gm, $k = 65000$ dynes/cm and $b = 70$ gm/s. What is the (i) damping angular frequency? (ii) damping frequency? (iii) damping period? (iv) natural frequency? (v) damping factor? (vi) damping amplitude? (vii) life time? (viii) displacement equation? (ix) its displacement at $t = 1$ sec? (x) its velocity at $t = 2$ sec? (xi) its acceleration at $t = 3$ sec? (xii) displacement of the particle after 1 cycle? and (xiii) damping acceleration of the particle after 3 cycles?
29. Graph illustrates the displacement vs. time graph for a mass spring system oscillating in Damped Harmonic Oscillation. The mass of the mass-spring system is 50 g. Damping co-efficient is $b = 50$ gm/s. Construct the equation of displacement vs. time. (i) Find out the damped angular frequency, (ii) Find out the damping frequency, (iii) Find out the damping energy $t = 0.5$ sec, (iv) Find out the damping time period, (v) Find out the spring constant, (vi) Find out the damping factor, (vii) Find out the damping term at $t = 0.2$ sec, (viii) Find out the relaxation time, (ix) Find out the displacement at $t = 2$ sec, (x) Calculate the velocity at $t = 2$ sec, (xi) Calculate the maximum velocity at $t = 2$ sec, (xii) Calculate the acceleration at $t = 2$ sec, (xiii) Calculate the maximum acceleration at $t = 4$ sec, (xiv) Determine the kinetic energy at $t = 0.5$ sec, (xv) Determine the potential energy at $t = 0.8$ sec, and (xvi) Determine the total energy at $t = 0.5$ sec.



30. Graph illustrates the displacement vs. time graph for a mass spring system oscillating in Damped Harmonic Oscillation. The mass of the mass-spring system is 50 g. Damping co-efficient is $b = 50$ gm/s. Construct the equation of displacement vs. time. Find out the (i) damping energy at $t = 0.5$ sec, (ii) displacement at $t = 1$ sec, (iii) velocity at $t = 2$ sec, (iii) acceleration at $t = 3$ sec, (iv) kinetic energy at $t = 4$ sec, (v) potential energy at $t = 5$ sec, and (vi) total energy at $t = 6$ sec.

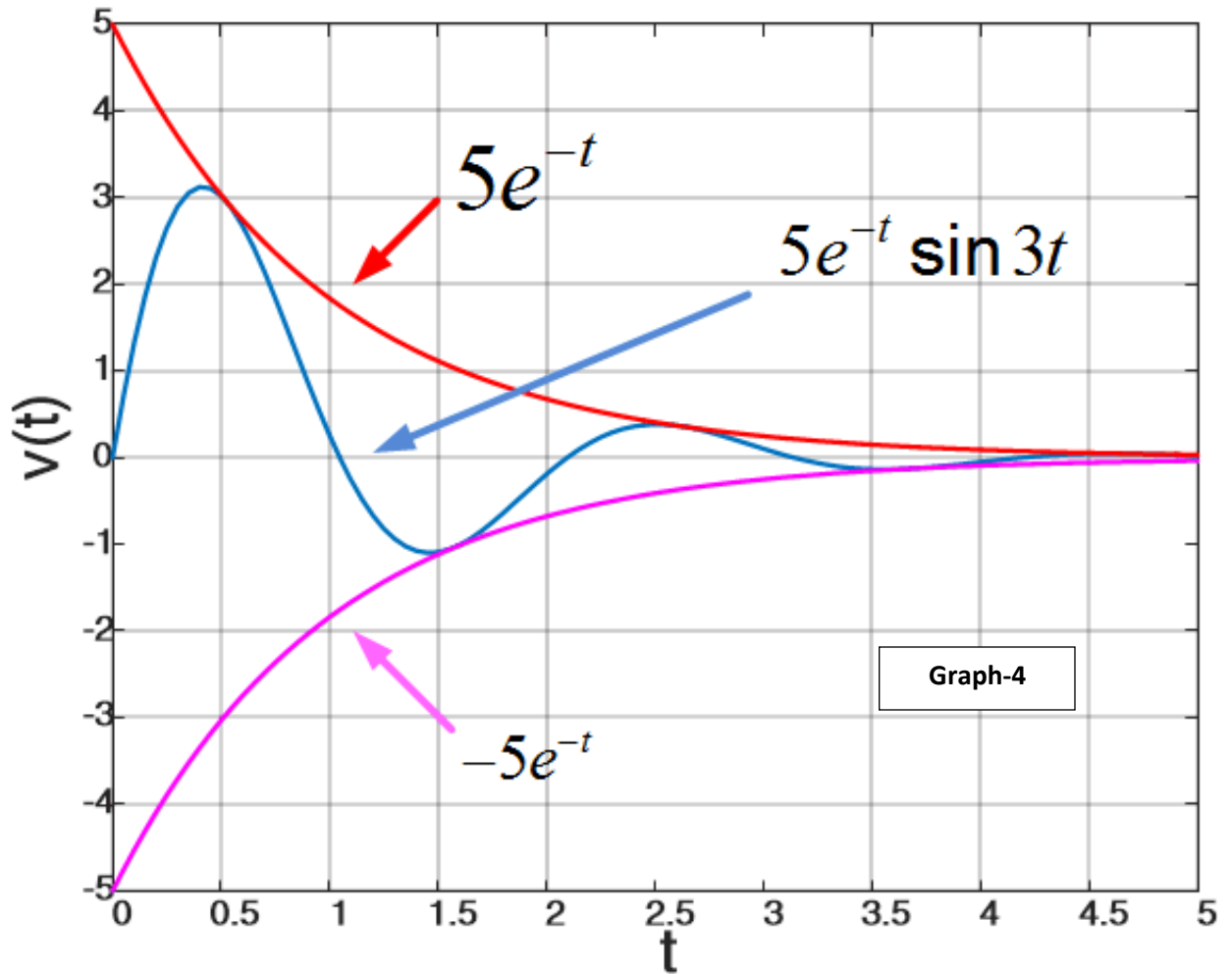


31. Graph illustrates the displacement vs. time graph for a mass spring system oscillating in Damped Harmonic Oscillation. The mass of the mass-spring system is 50 g. Damping co-efficient is $b = 50$ gm/s. Construct the equation of displacement vs. time. Find out the (i) damping energy at $t = 0.5$ sec, (ii) displacement at $t = 1$ sec, (iii) velocity at $t = 2$ sec, (iii) acceleration at $t = 3$ sec, (iv) kinetic energy at $t = 4$ sec, (v) potential energy at $t = 5$ sec, and (vi) damping amplitude at $t = 6$ sec.

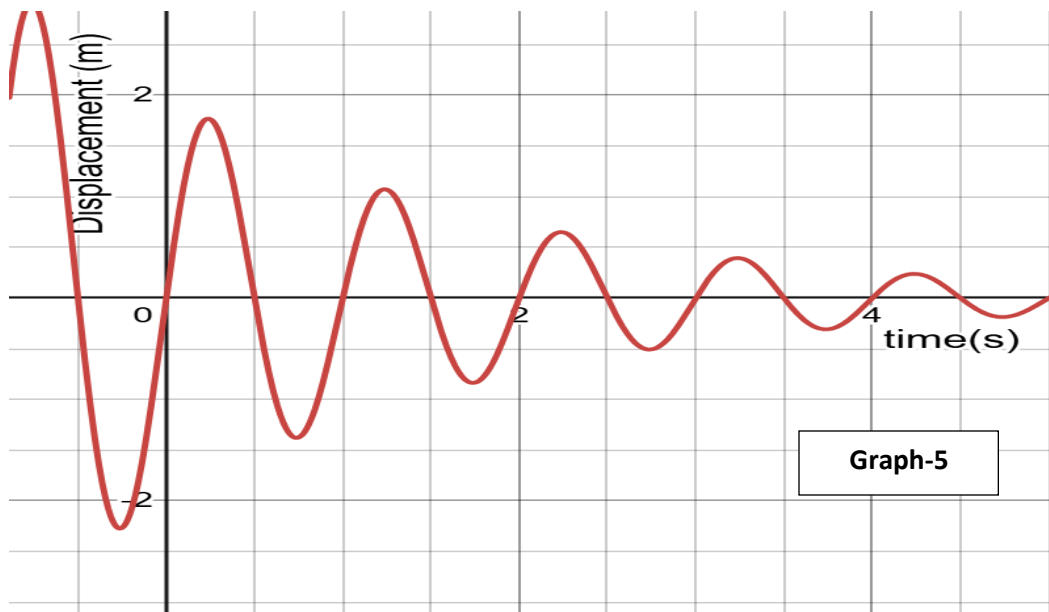


32. A typical DHM is generated due to the frictional motion of a spring of mass 250 gm. The displacement profile is shown below. Find out the (i) damping amplitude at $t = 2$ sec, (ii) damping

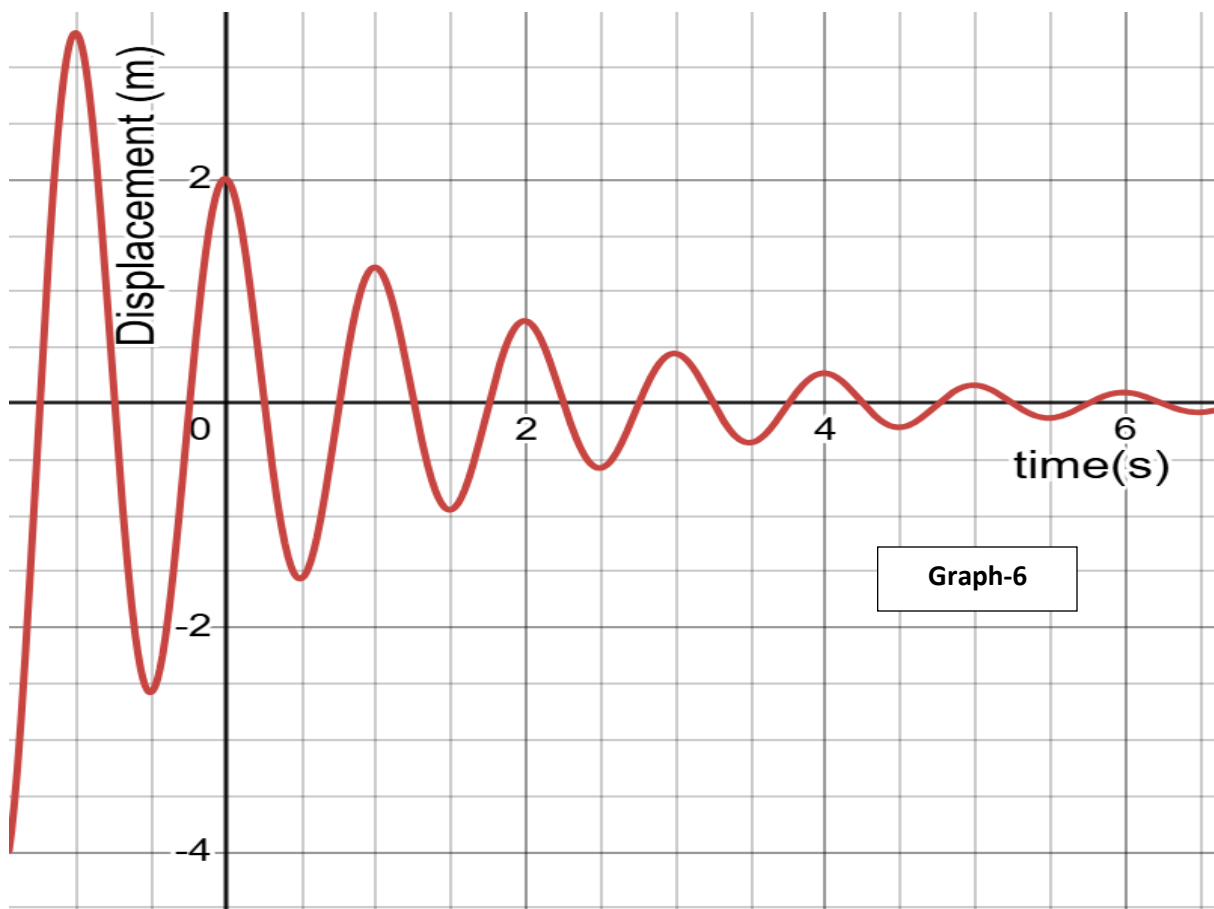
displacement at $t=3\text{sec}$, (iii) damped angular frequency, (iv) damping energy at $t=4\text{ sec}$, (v) maximum velocity at $t=5\text{ sec}$, (vi) maximum acceleration at $t=6\text{ sec}$, (vii) life time of the DHM, (viii) damping coefficient, (ix) decay envelope $e^{-\alpha t}$, at $t=7\text{ sec}$, and (x) envelope of DHM.



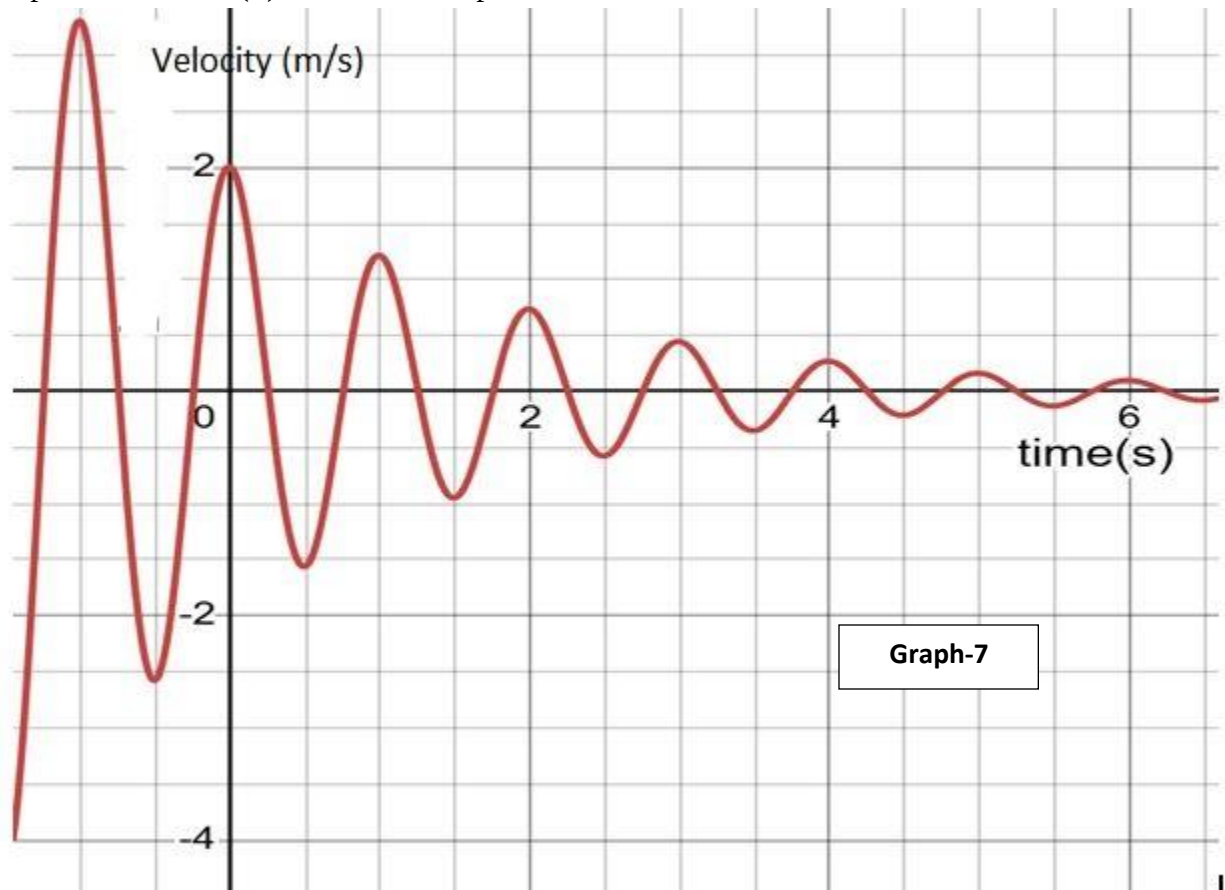
33. A typical DHM is generated due to the frictional motion of a spring of mass $m=1\text{ kg}$. The displacement profile is shown below where $b=1\text{ kg/s}$. From the graph, find out the (i) equation of displacement, (ii) damping amplitude A_d at $t=1.25\text{ sec}$, (iii) damping displacement $y = y_d$, at $t=3\text{sec}$, (iii) damped angular frequency ω_d , (iv) damping energy at $t=3.25\text{ sec}$, (v) maximum velocity, (vi) maximum acceleration, (vii) life time of the DHM, and (viii) damping envelope $\pm Ae^{-\frac{b}{2m}t}$, at $t=2.5\text{s}$.



34. A typical DHM is generated due to the frictional motion of a spring of mass $m=1$ kg, damping coefficient $b=1$ kg/s. (i) What is the equation of displacement? Also find out the (ii) damping amplitude, damping displacement, damping frequency, and damping energy, at $t=1$ sec, and (iii) relaxation time of the DHM.



35. A typical DHM is generated due to the frictional motion of a spring of mass $m=500$ g, damping factor $\alpha=0.5$ s⁻¹. The velocity equation is represented below. (i) What is the equation of displacement? and (ii) What is the displacement at $t=3$ s?



36. A typical displacement vs time graph for a damped oscillatory system shown below which is generated due to the frictional motion of a spring of mass $m=1200$ g, spring constant $k=65$ N/m. (i) Calculate the value of b , (ii) life time of oscillations, (iii) equation of displacement, and (iii) displacement at $t=2.5$ s.

